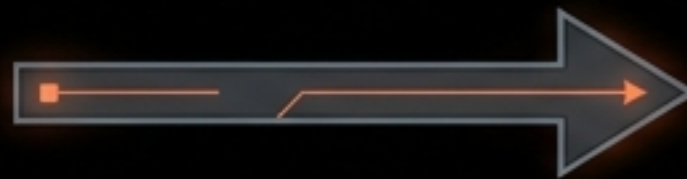


Thermal Image 8x Super-Resolution

Deep Learning, Spectral Supervision, Edge Injection, and Physics-Aware Reconstruction

The Core Problem: A 64:1 Information Deficit



8× super-resolution is an extreme inverse problem.
It is not making an image bigger.

98.4% of high-frequency spatial information is
irreversibly destroyed during downsampling.

The network does not recover lost information—it hallucinates based on learned priors.

The Domain Mismatch Matrix

NATURAL RGB IMAGES

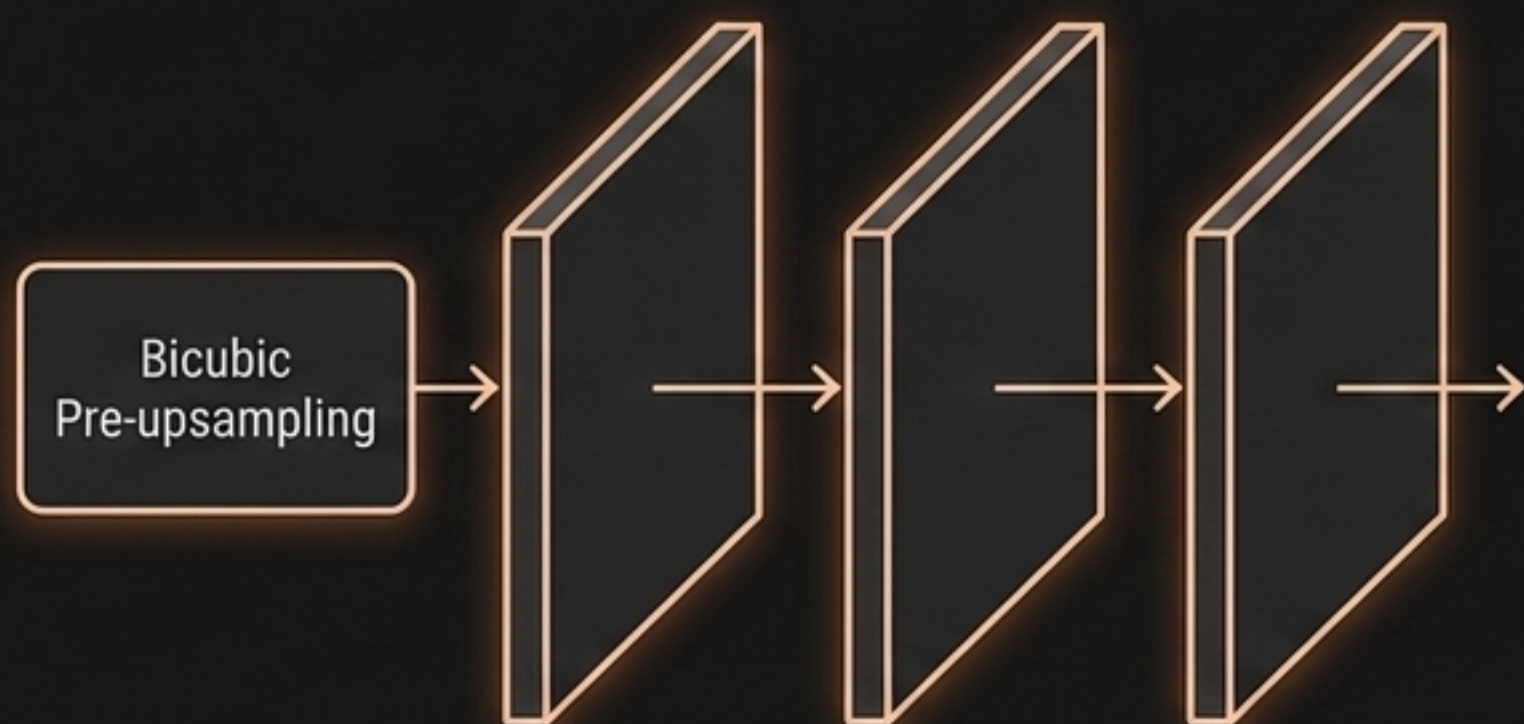
- Property: Texture-heavy.
- Structure: Stochastic micro-patterns.
- Frequency: Broad spectrum distribution.
- Network Behavior: Optimizes for visual realism.

THERMAL IMAGES

- Property: Field-heavy.
- Structure: Smooth heat diffusion & sharp thermodynamic boundaries.
- Frequency: Steep low-frequency power laws.
- Network Behavior: Demands physical truth.

A model trained to hallucinate natural textures will invent fake physics when applied to thermal data.

Phase 1: The Shallow Baseline & The Bicubic Trap

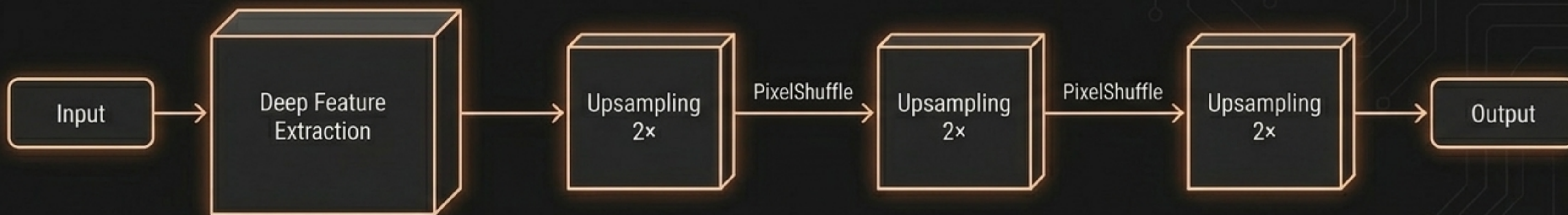


Models Evaluated: SRCNN (MSE) & SRCNN (L1).

Observation: Bicubic pre-upsampling destroys edge topology before the network even sees the data.

Result: Severe oversmoothing. The L1 median is slightly sharper than the MSE mean, but both entirely fail to resolve the extreme 8× scale.

Phase 2: Progressive Upsampling & The Diffusive Prior



Motivation: Eliminate bicubic pre-upsampling by processing deep features at low-resolution and upsampling progressively.

Original Contribution: The Diffusive Patch Loss. Penalizing local mean variance to enforce smooth, noise-free isothermal heat regions.



Phase 3: Deep Residual Hierarchies

Motivation: 16 residual blocks to capture long-range isothermal dependencies.

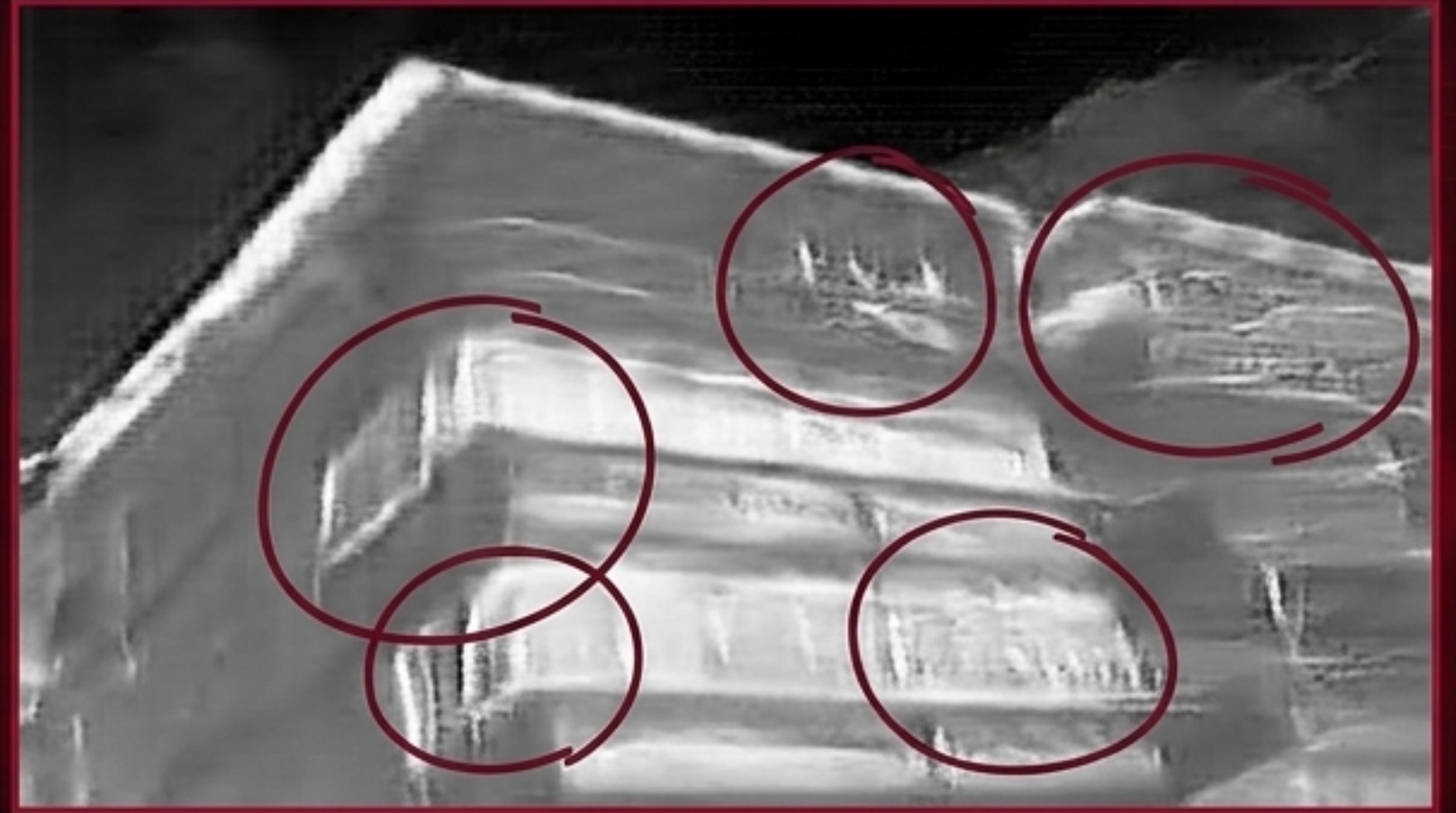
Evolution: Upgrading to Gradient-Domain Diffusive Loss.

Core Insight: Instead of matching isolated pixels, the network is now matching the thermodynamic heat-flux field (spatial gradients).

Evolution HUD

Model 4: SRResNet Hybrid

The VGG Contradiction & The Hallucination Crisis



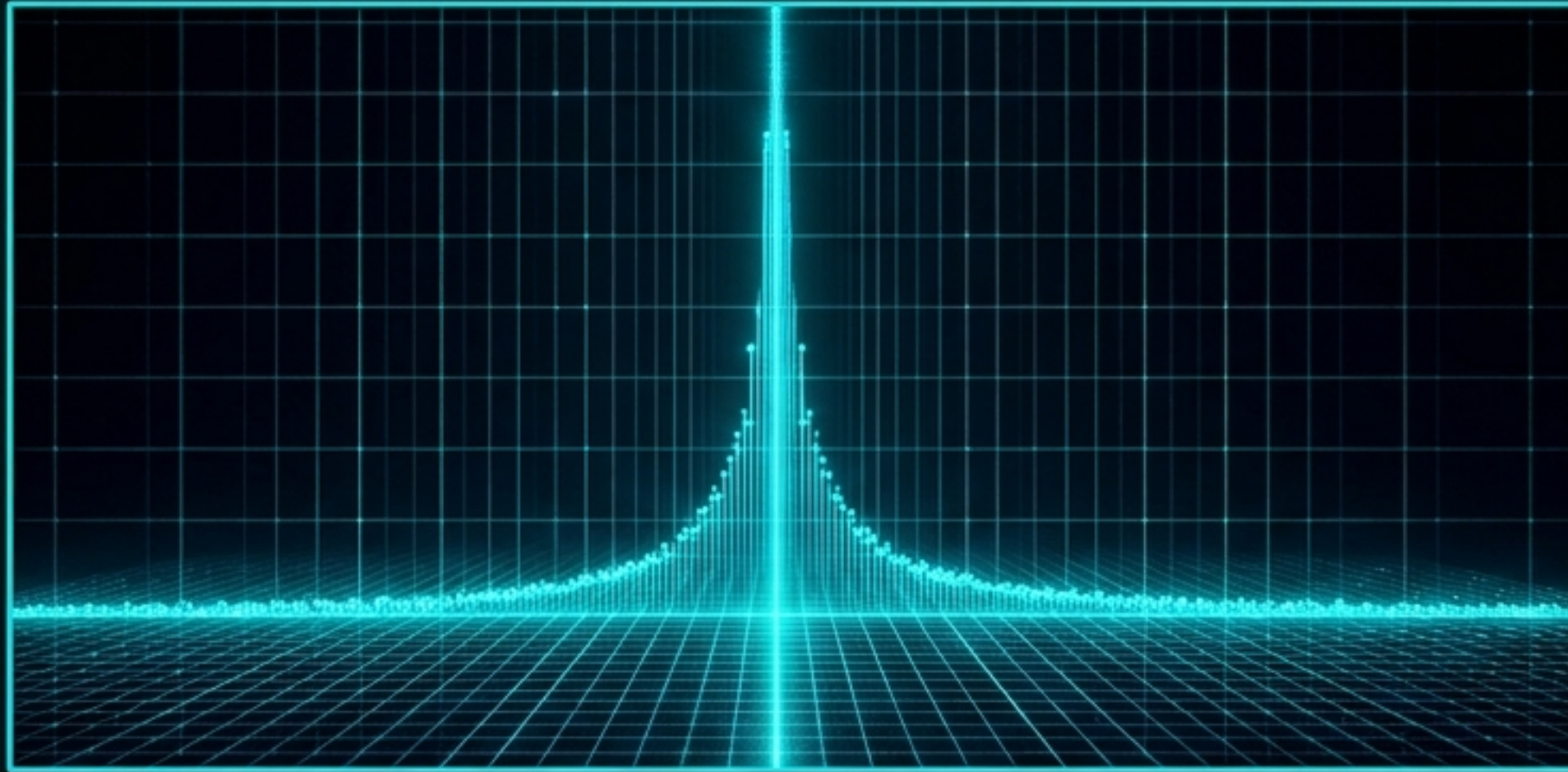
The Crisis: Adversarial training introduces sharpness, but VGG perceptual loss is trained on natural images.

The Research Insight: GANs optimize for statistical realism, not physical truth. The model hallucinates natural textures into smooth thermal fields.

The Danger: A false texture in thermal imaging implies a non-existent heat fluctuation.

**Thermal reconstruction is
not about visual sharpness.
It is about thermodynamic
topology and spectral fidelity.**

Phase 4: Spectral Supervision (The FFT Breakthrough)



Hypothesis: Hallucinated textures create unnatural high-frequency energy spikes.

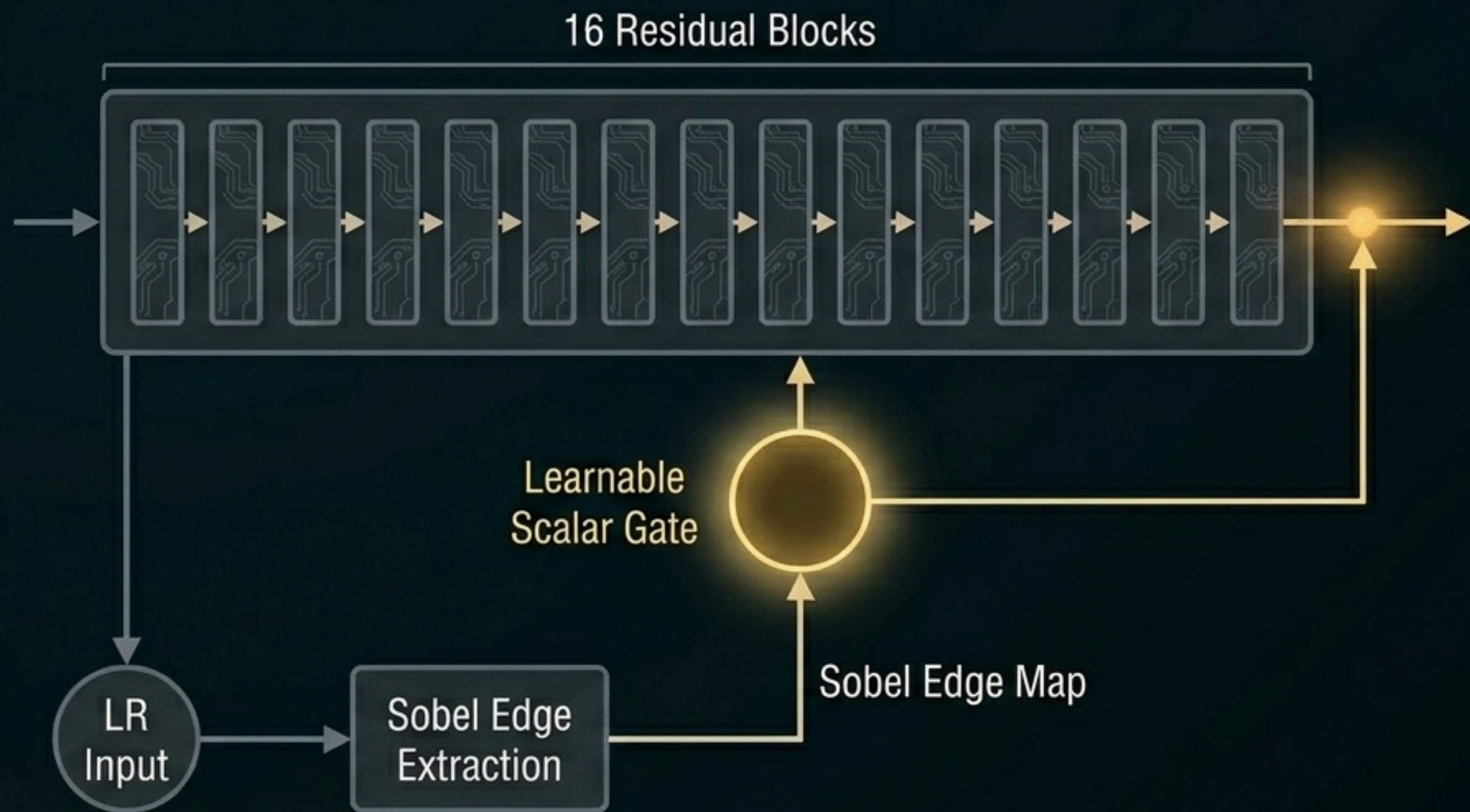
Contribution: FFTLoss. An explicit L1 penalty on the Fourier magnitude spectrum.

Result: The network is forced to obey the steep low-frequency dominance of true thermal fields. False textures are suppressed; spectral realism is achieved.

Evolution HUD timeline is updated:



Phase 5: Architectural Inductive Bias

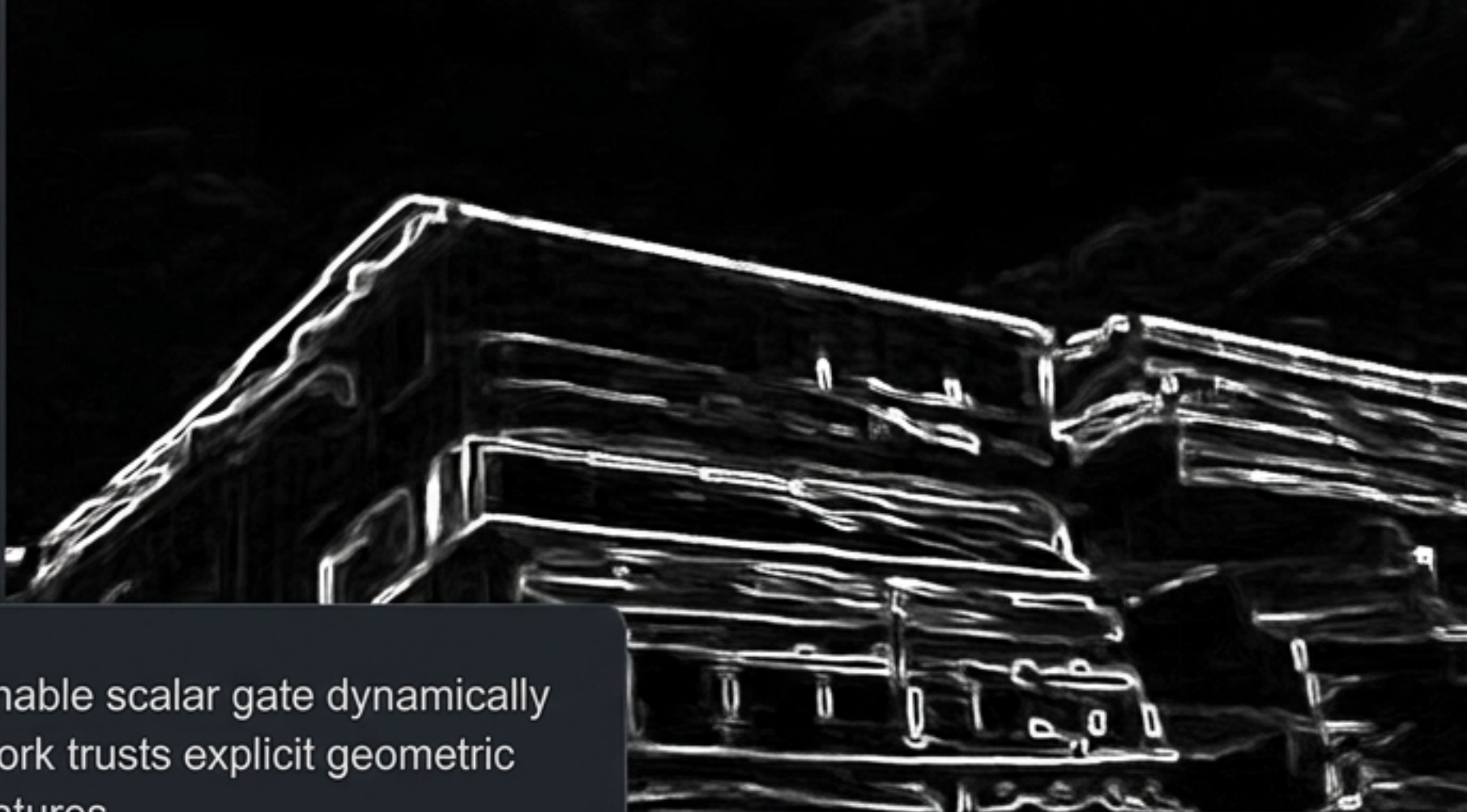


- **Observation:** Even at extreme 8× downsampling, low-resolution edge topology survives.
- **Contribution:** Extracting Sobel edge maps from LR input and injecting them deep into the residual feature stream.
- **Inductive Bias:** Boundaries are physically anchored, bypassing the network's need to rediscover them.

Evolution HUD timeline is updated:

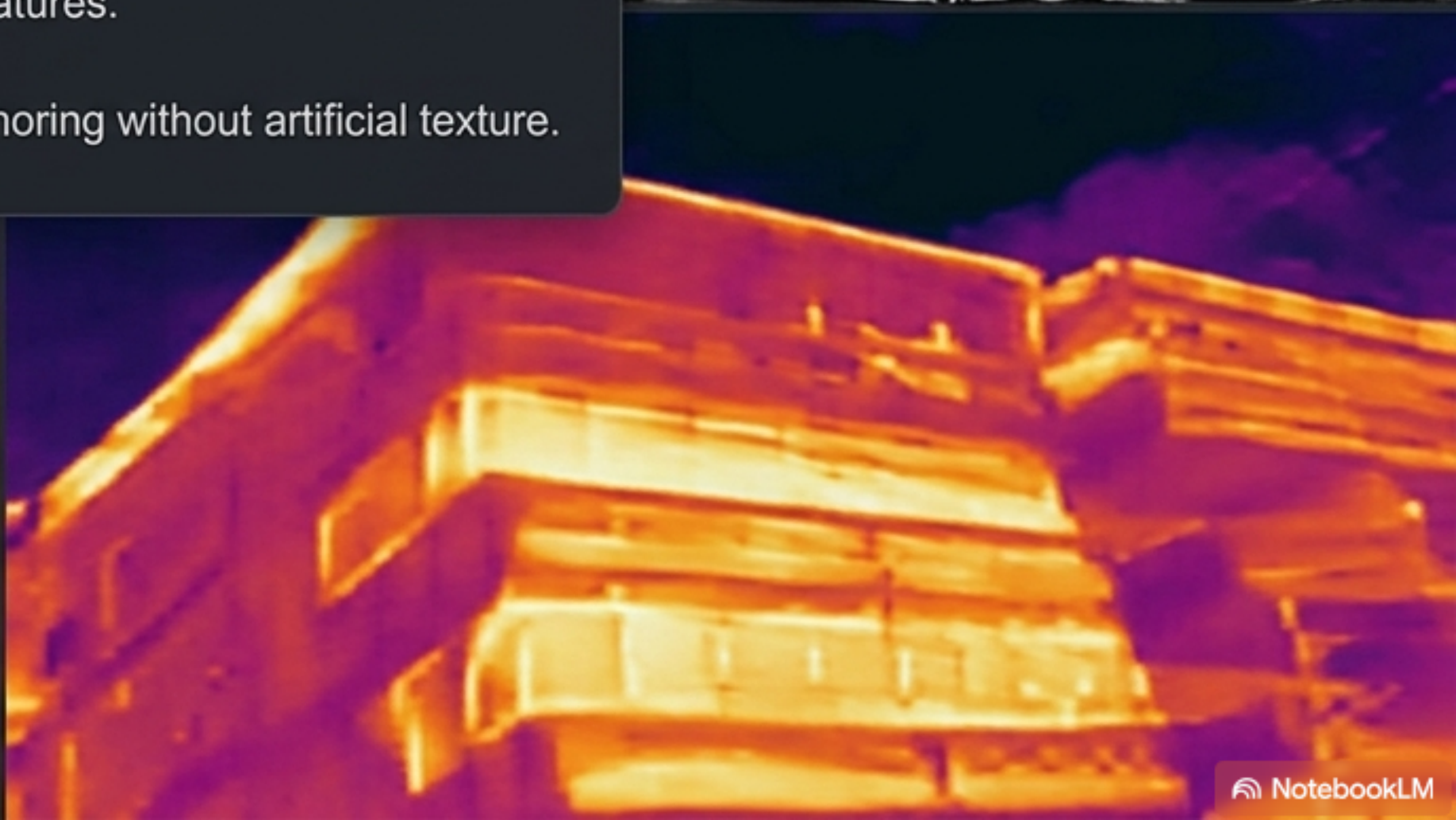
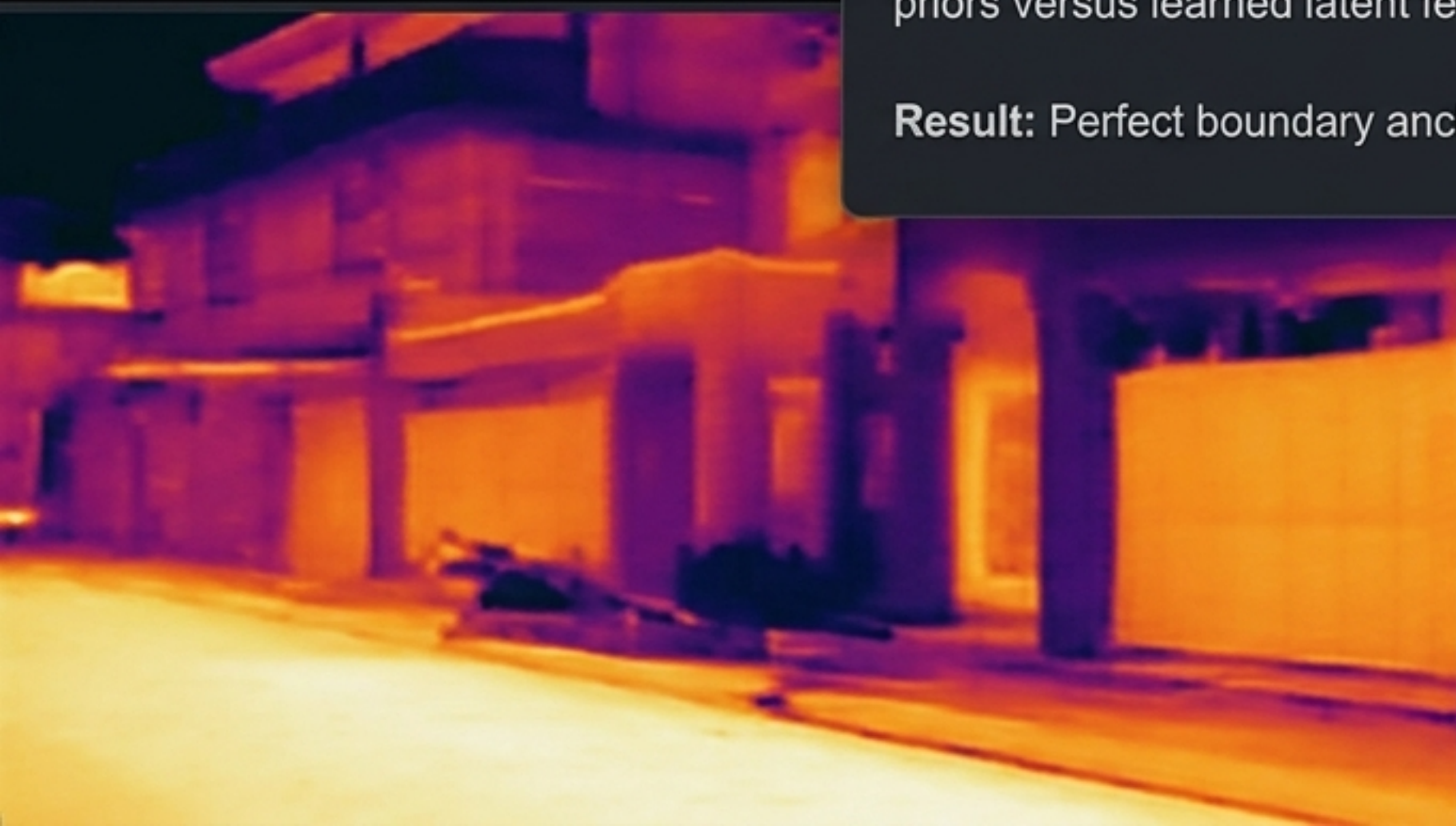
Model 6: Edge Injection SRResNet

Visualizing Geometric Inductive Bias

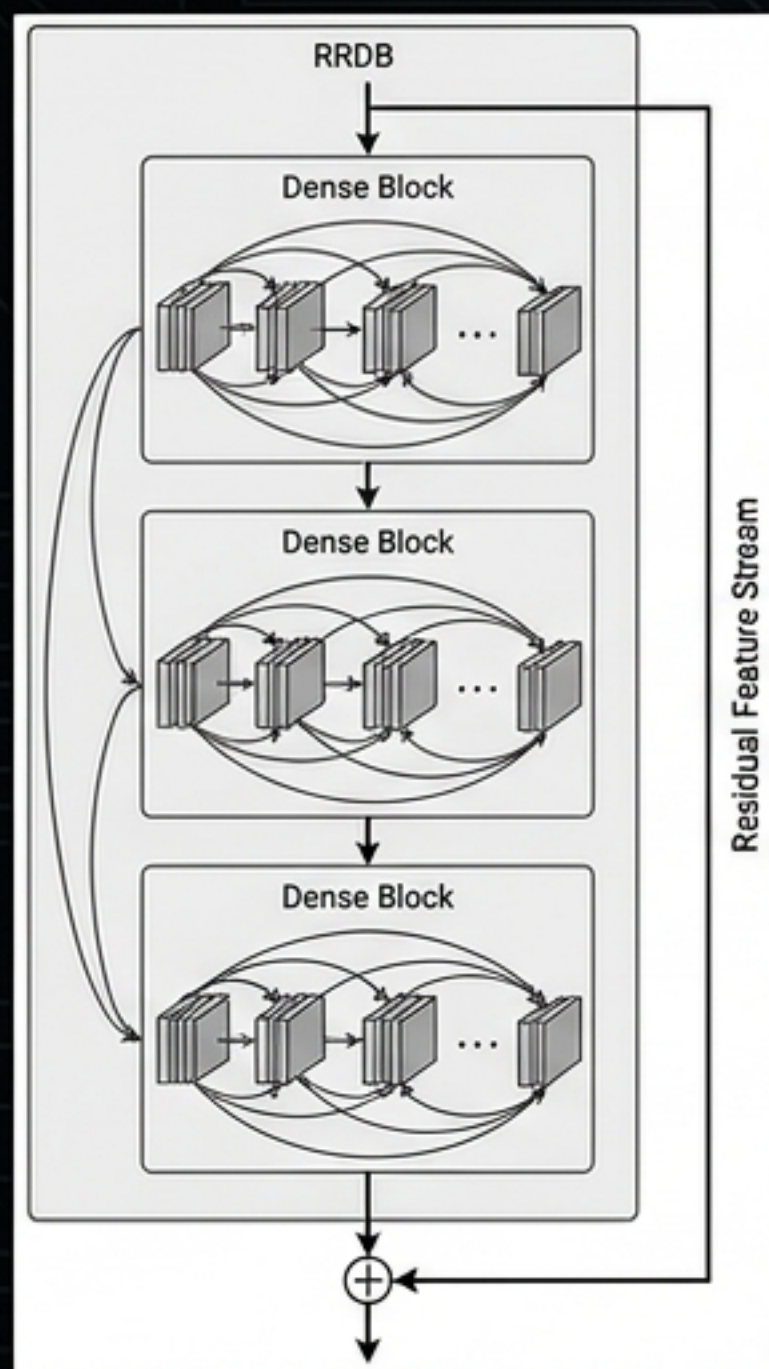


Mechanism Insight: The learnable scalar gate dynamically calculates how much the network trusts explicit geometric priors versus learned latent features.

Result: Perfect boundary anchoring without artificial texture.



Phase 6: Stabilized Adversarial Networks



Motivation: Harnessing GAN sharpness without the prior hallucination instability.

Implementation Stack:

- **Spectral Normalization:** Constrains Lipschitz bounds.
- **Feature Matching:** Enforces multi-scale structural stability.
- **RaGAN:** Relativistic distribution learning for comparative realism.



Evolution HUD timeline is updated:

Phase 7: The Magnum Opus

$$\begin{aligned} \mathcal{L}_{L1} = & \frac{1}{2} \sum_{i=0}^N | \cdot | + \underbrace{\mathcal{V}_{VGG}(\psi)}_{\text{Median Fidelity}} + \underbrace{\mathcal{L}_{\text{diffusive}}}_{\text{Structural Prior}} + \underbrace{\mathcal{L}_{\text{FFT}} \otimes [\mathbf{W}(H_{o,ot}^2) \gamma_j]}_{\text{Heat Flux Mapping}} \\ & + \frac{1}{2} \sum_{i=0}^N | \cdot | + \underbrace{\text{Feature Matching}}_{\text{Spectral Topology}} + \underbrace{\mathcal{L}_{\text{RaGAN}}}_{\text{Scale Stability}} + \underbrace{R1_{\text{pen}} - R_{\mu}}_{\text{Discriminator Smoothness}} \end{aligned}$$

Stabilization Note: A rigorous 3-phase warmup schedule is utilized to prevent early adversarial collapse.

Evolution HUD timeline is updated:

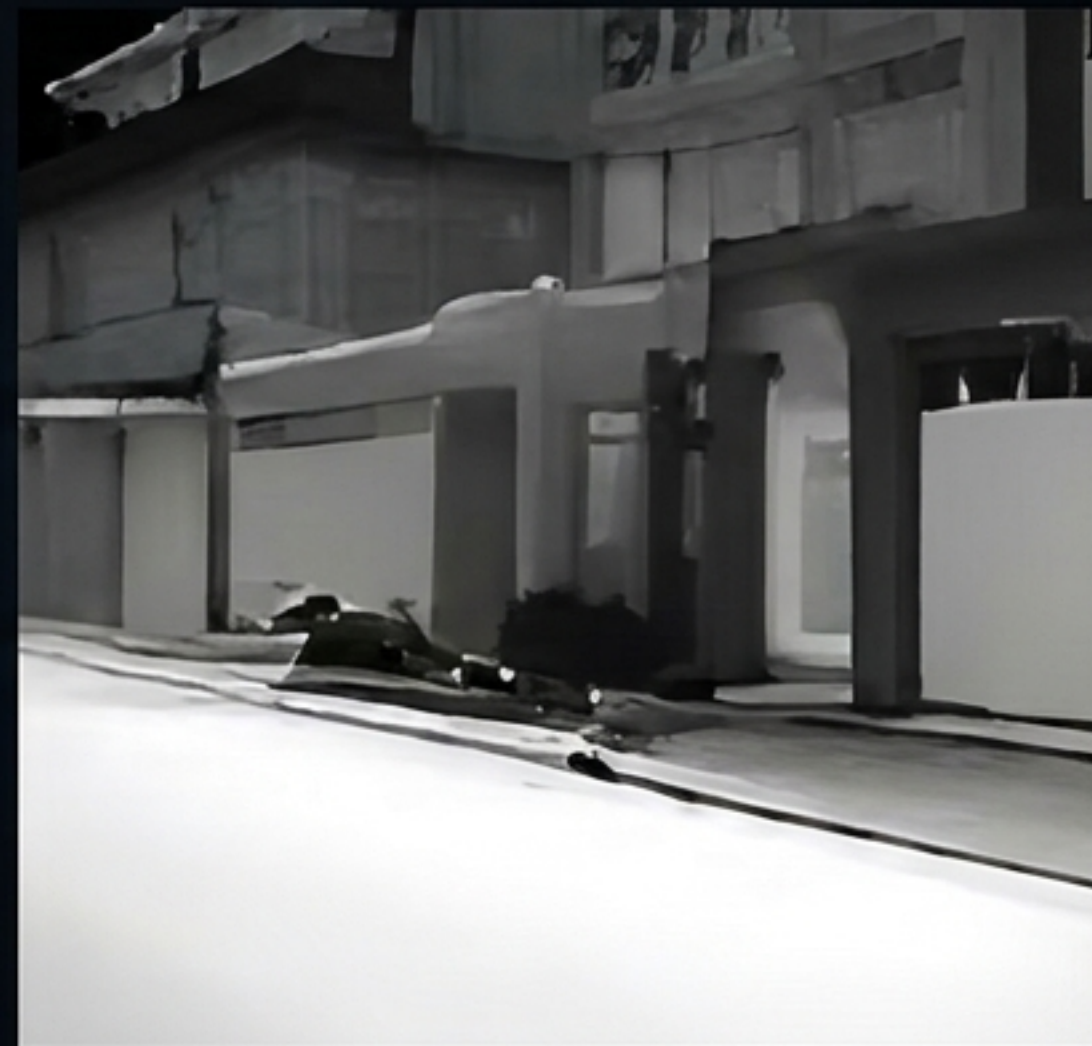
The 8× Validation Benchmark – Scene A (Street Geometry)



SRCNN
(Early L1 Oversmoothing)



SRGAN
(Adversarial Hallucination)



Thermal Real-ESRGAN
(Physics-Aware Synthesis)

The 8× Validation Benchmark – Scene B (Facade Topologies)



SRCNN
(Shallow Baseline)



Sobel
Boundary Extraction



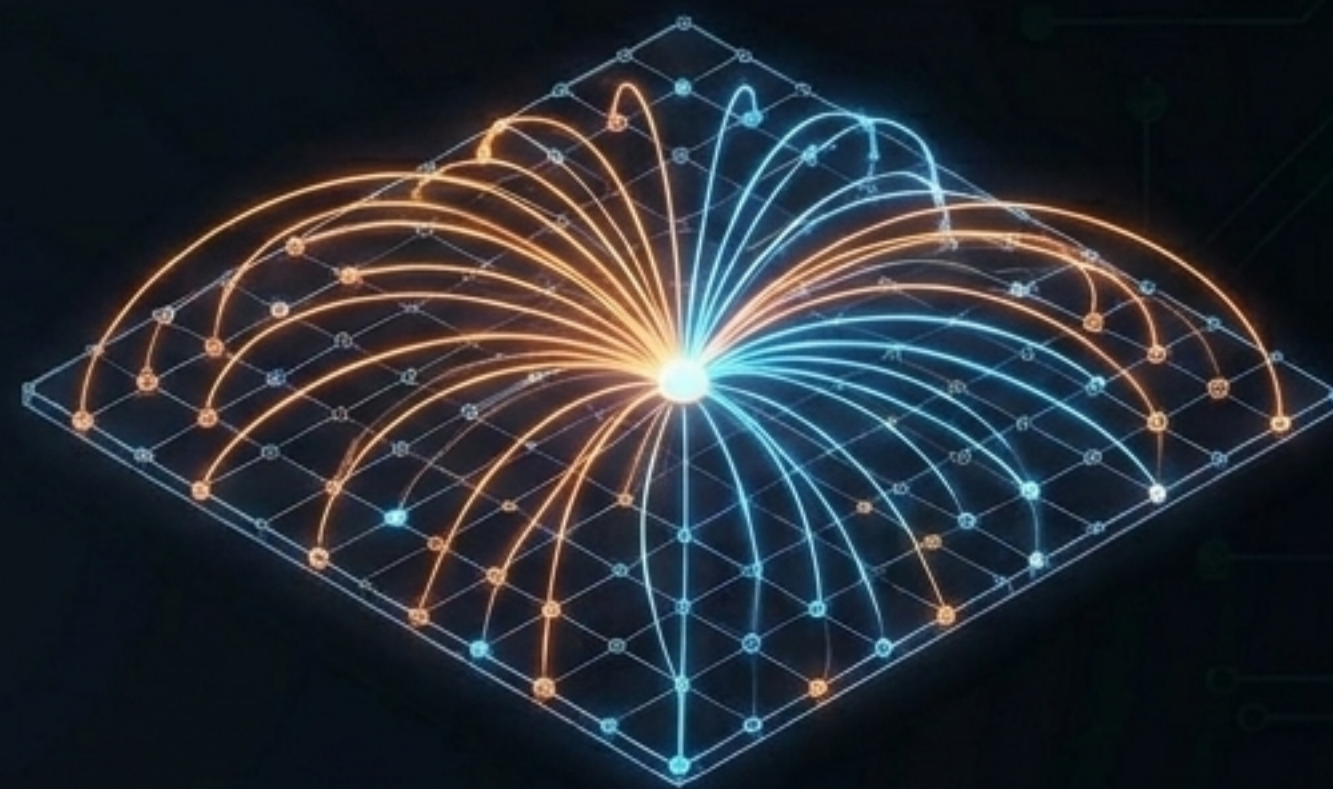
Thermal Real-ESRGAN
(Final Dense-Block Reconstruction)

The Attention Curiosity: What Convolution Misses



Finite Receptive Field
(Convolution)

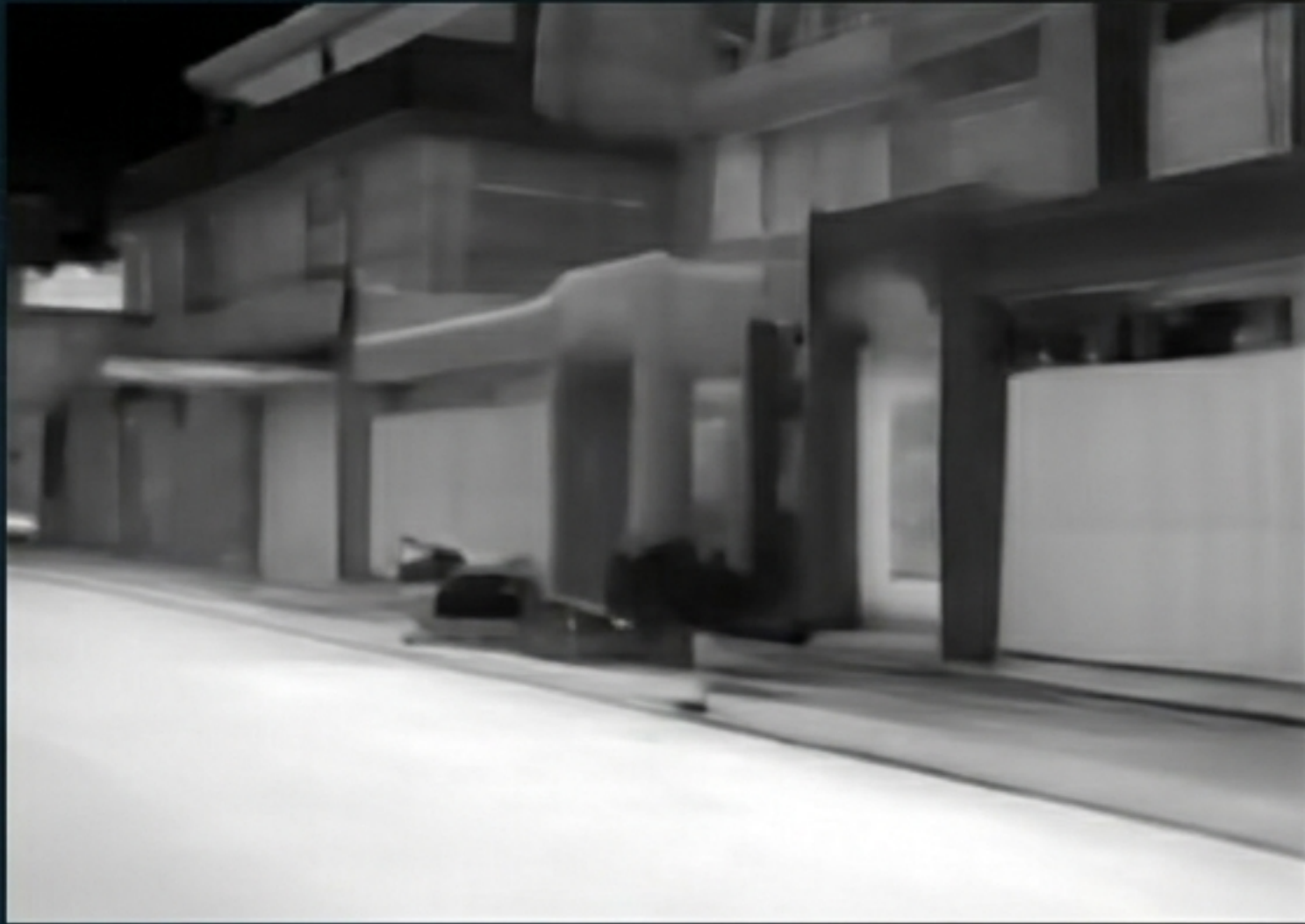
The Limitation: Even with 16 ResBlocks, receptive fields remain finite. Thermal gradients, however, span entire images.



Global Attention Mechanism
(Transformers)

The Hypothesis: Transformers (Shifted Window Attention) can theoretically solve long-range isothermal dependencies that convolutions cannot reach.

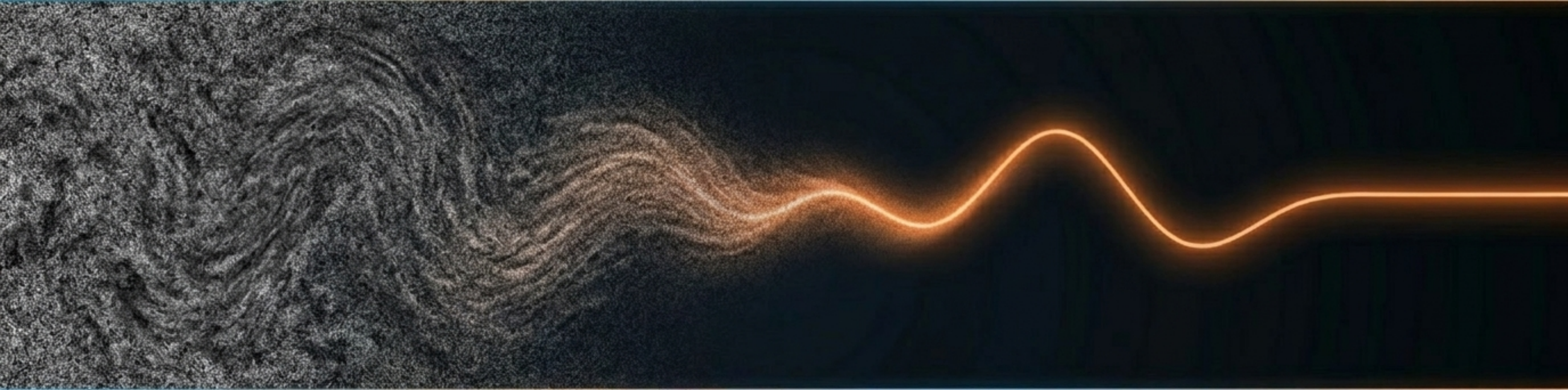
Phase 8: SwinIR Experiments



Observation: Attention mechanisms successfully integrate global thermal context.

Conclusion: Proves that non-local processing is highly effective for maintaining isothermal continuity across vast smooth spaces, albeit with extreme computational costs.

Phase 9: The Stochastic Horizon



The Next Frontier: Deterministic models collapse a 64:1 uncertainty deficit into a single point. Thermal reality, however, is **probabilistically** uncertain.

The Vision: Denoising Diffusion Probabilistic Models (DDPM) can map the full posterior distribution of thermal topologies, providing essential uncertainty maps critical for safety and inspection.

Model 11: The Stochastic Horizon (Diffusion)

Original Research Contributions

- 1** Gradient-Domain Diffusive Loss (Preserving Heat Flux Priors)
- 2** FFT Spectral Loss (Enforcing Low-Frequency Thermodynamic Dominance)
- 3** Learnable Sobel Edge Injection Architecture (Geometric Anchor Constraints)
- 4** Conservative 3-Phase Adversarial Thermal Stabilization (Eliminating Hallucinations)

**Super-resolution in the thermal domain
is not an act of image processing.
It is an act of physical archaeology—extracting
thermodynamic truth from collapsed signals.**